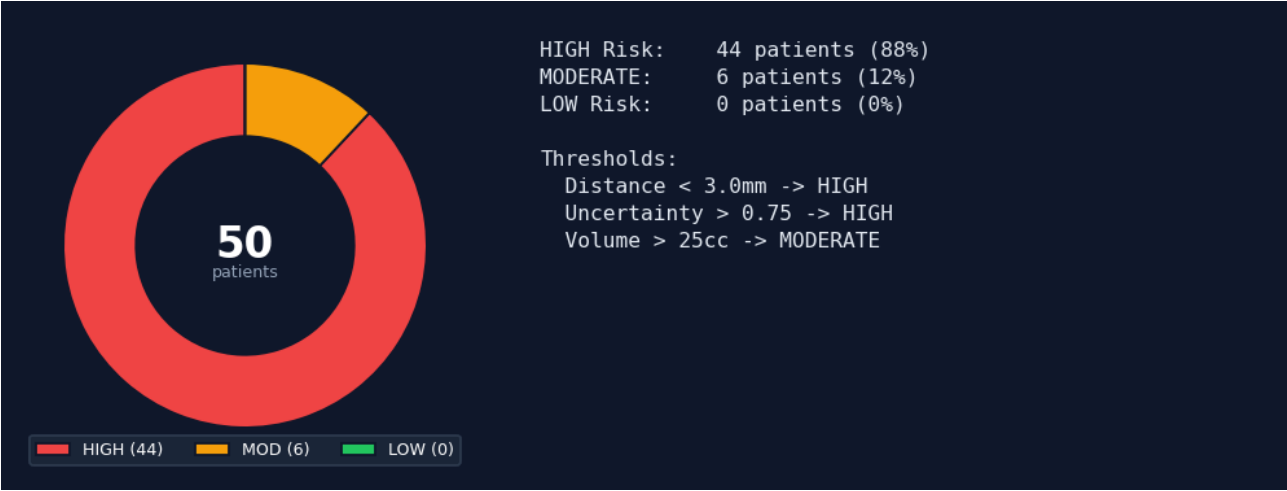


NeuroQA Copilot - Group Report

Cohort: 50 patients | BraTS 2020 Dataset | Generated: 2026-06-05 15:11

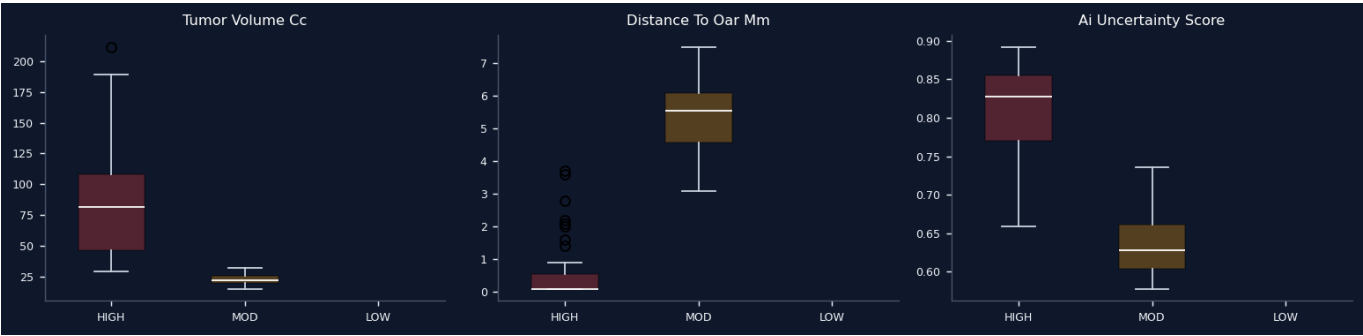
Risk Distribution



Descriptive Statistics

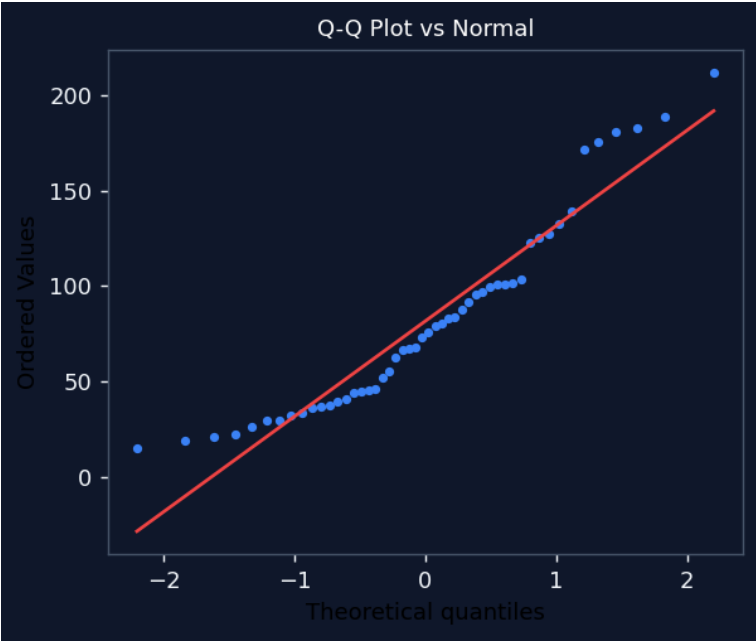
Metric	N	Mean	Median	Std	Min	Max	Skew	Kurt	CV%
Tumor_Volume_cc	50	81.65	74.69	0.00	14.83	211.98	0.86	0.02	0.0
Distance_to_OAR_mm	50	1.15	0.10	0.00	0.10	7.50	1.91	2.87	0.0
AI_Uncertainty_Score	50	0.79	0.81	0.00	0.58	0.89	-1.11	0.67	0.0
Irregularity_Score	50	0.44	0.40	0.00	0.28	0.84	1.13	0.48	0.0
Max_Diameter_mm	50	81.69	80.85	0.00	48.30	127.90	0.48	0.30	0.0

Distributions by Risk Level

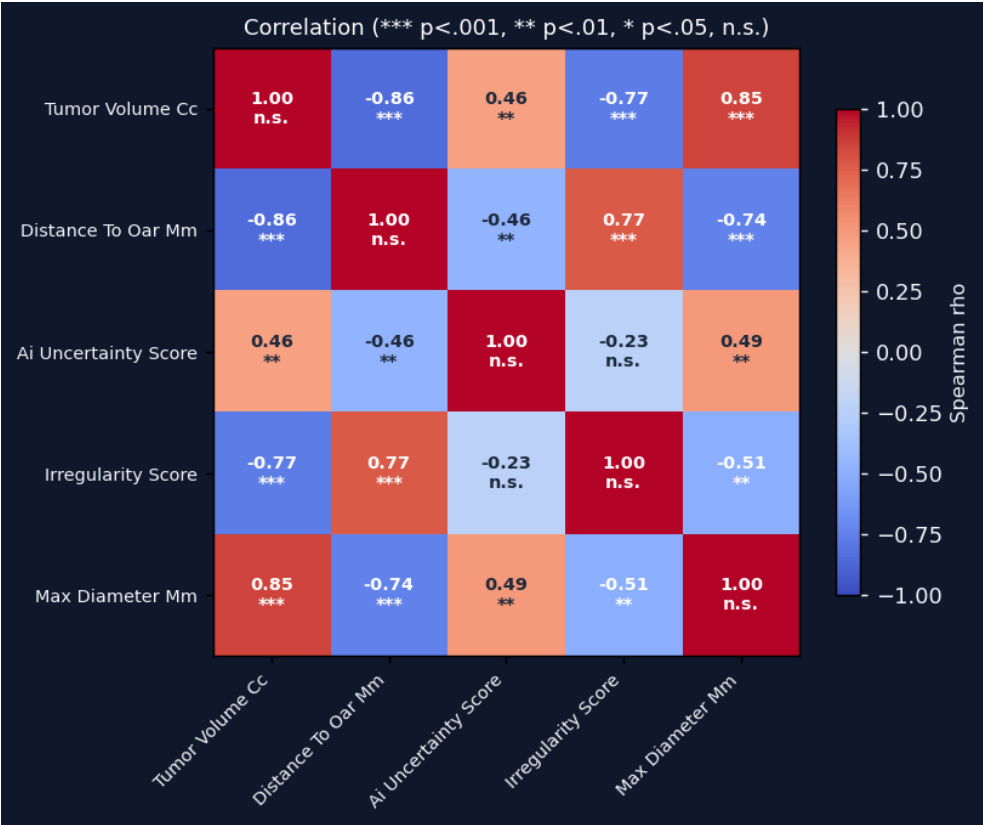


Distribution Fitting (Tumor Volume)

Distribution	AIC	KS Stat	KS p-value	Best
Normal	537.5	0.0000	0.0000	
Log-Normal	524.5	0.0000	0.0000	
Gamma	523.8	0.0000	0.0000	Yes

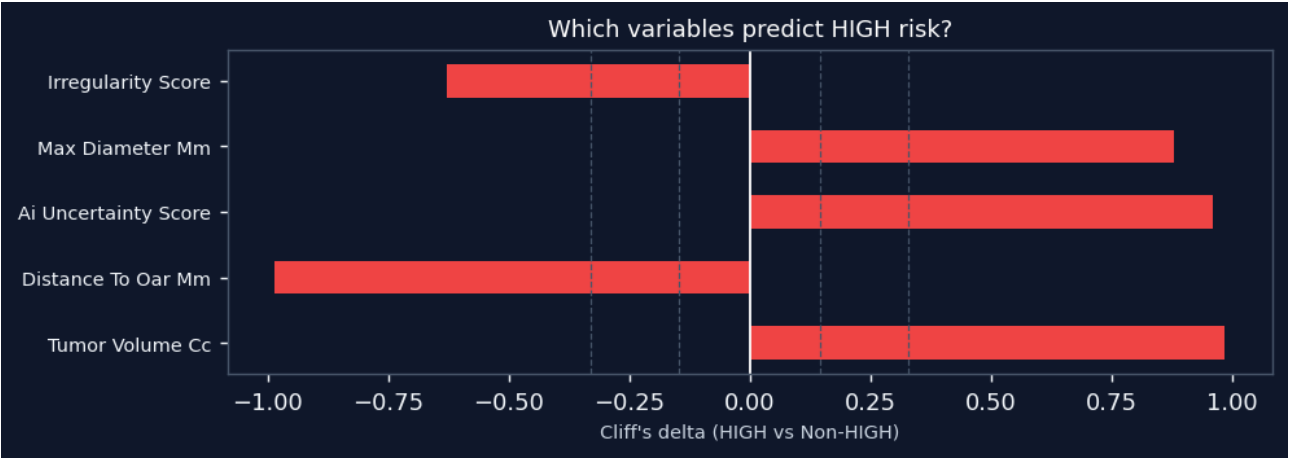


Spearman Correlation Matrix



Risk Factor Analysis (HIGH vs Others)

Variable	Mann-Whitney U	p-value	Cliff's d	Effect	AUC
Tumor Volume cc	0.0	0.0000	0.985	Large	0.992
Distance to OAR mm	0.0	0.0000	-0.985	Large	0.008
AI Uncertainty Score	0.0	0.0002	0.962	Large	0.981
Max Diameter mm	0.0	0.0006	0.879	Large	0.939
Irregularity Score	0.0	0.0138	-0.629	Large	0.186



Group Comparisons (Kruskal-Wallis)

Tumor Volume Cc

H statistic: 15.062
p-value: 0.000104
eta-squared: 0.2930
Verdict: Significant

Distance To Oar Mm

H statistic: 20.410
p-value: 0.000006
eta-squared: 0.4044
Verdict: Significant

Ai Uncertainty Score

H statistic: 14.378
p-value: 0.000150
eta-squared: 0.2787
Verdict: Significant

Bootstrap 95% Confidence Intervals (10K resamples)

Variable	Observed	95% CI Lower	95% CI Upper
Tumor Volume Cc	81.651	68.264	95.760
Distance To Oar Mm	1.154	0.682	1.708
Ai Uncertainty Score	0.791	0.769	0.811

Outlier Detection (IQR Method)

Variable	N	Outliers (IQR)	% Outliers	Low	High
Tumor_Volume_cc	50	1	2.0	0	1
Distance_to_OAR_mm	50	5	10.0	0	5
AI_Uncertainty_Score	50	2	4.0	2	0
Irregularity_Score	50	3	6.0	0	3
Max_Diameter_mm	50	1	2.0	0	1

Complete Patient List

Patient ID	Structure	Volume (cc)	Dist OAR (mm)	Uncertainty	Risk
BraTS20_Training_001	Chiasm	212.0	0.1	0.848	HIGH
BraTS20_Training_026	Left Optic Nerve	180.6	0.1	0.755	HIGH
BraTS20_Training_027	Right Retina	66.7	0.1	0.811	HIGH
BraTS20_Training_029	Hippocampus	62.4	0.1	0.827	HIGH
BraTS20_Training_030	Right Retina	99.9	0.1	0.864	HIGH
BraTS20_Training_031	Cochlea (Right)	87.6	0.1	0.880	HIGH
BraTS20_Training_033	Right Optic Nerve	122.7	0.1	0.710	HIGH
BraTS20_Training_034	Brainstem	91.6	0.1	0.828	HIGH
BraTS20_Training_035	Right Optic Nerve	78.8	0.1	0.851	HIGH
BraTS20_Training_037	Right Retina	189.0	0.1	0.781	HIGH
BraTS20_Training_038	Right Optic Nerve	100.9	0.1	0.820	HIGH
BraTS20_Training_039	Chiasm	51.9	0.1	0.785	HIGH
BraTS20_Training_040	Cochlea (Left)	171.4	0.1	0.829	HIGH
BraTS20_Training_041	Pituitary Gland	101.4	0.1	0.856	HIGH
BraTS20_Training_042	Left Retina	76.0	0.1	0.722	HIGH
BraTS20_Training_043	Optic Chiasm	37.4	2.0	0.829	HIGH
BraTS20_Training_045	Left Optic Nerve	97.2	0.1	0.767	HIGH
BraTS20_Training_046	Hippocampus	36.3	2.2	0.876	HIGH
BraTS20_Training_047	Chiasm	132.7	0.1	0.892	HIGH
BraTS20_Training_048	Left Optic Nerve	29.2	3.7	0.804	HIGH
BraTS20_Training_049	Left Retina	83.4	0.1	0.715	HIGH
BraTS20_Training_024	Brainstem	95.6	0.1	0.861	HIGH
BraTS20_Training_025	Right Optic Nerve	73.4	0.1	0.766	HIGH
BraTS20_Training_022	Pituitary Gland	68.1	0.1	0.871	HIGH
BraTS20_Training_002	Pituitary Gland	67.0	0.1	0.843	HIGH
BraTS20_Training_003	Left Retina	29.8	3.6	0.761	HIGH
BraTS20_Training_004	Right Optic Nerve	103.5	0.1	0.843	HIGH
BraTS20_Training_006	Brainstem	139.2	0.1	0.858	HIGH
BraTS20_Training_007	Right Optic Nerve	39.8	1.6	0.834	HIGH
BraTS20_Training_008	Hippocampus	33.5	2.8	0.659	HIGH
BraTS20_Training_009	Left Optic Nerve	183.1	0.1	0.854	HIGH
BraTS20_Training_010	Cochlea (Right)	44.6	0.8	0.762	HIGH
BraTS20_Training_011	Pituitary Gland	55.1	0.1	0.748	HIGH
BraTS20_Training_023	Right Optic Nerve	41.1	1.4	0.799	HIGH
BraTS20_Training_050	Pituitary Gland	82.8	0.1	0.771	HIGH
BraTS20_Training_014	Right Retina	80.4	0.1	0.886	HIGH
BraTS20_Training_015	Chiasm	125.3	0.1	0.858	HIGH
BraTS20_Training_016	Right Optic Nerve	127.2	0.1	0.796	HIGH
BraTS20_Training_017	Hippocampus	100.8	0.1	0.827	HIGH
BraTS20_Training_018	Right Lens	45.1	0.7	0.808	HIGH
BraTS20_Training_019	Right Lens	43.8	0.9	0.859	HIGH
BraTS20_Training_020	Left Optic Nerve	175.7	0.1	0.833	HIGH
BraTS20_Training_021	Chiasm	36.9	2.1	0.815	HIGH
BraTS20_Training_013	Hippocampus	46.3	0.5	0.749	HIGH
BraTS20_Training_036	Brainstem	14.8	7.5	0.578	MODERATE
BraTS20_Training_032	Right Lens	26.5	4.3	0.639	MODERATE
BraTS20_Training_028	Left Retina	20.9	5.7	0.600	MODERATE
BraTS20_Training_044	Right Retina	19.1	6.2	0.668	MODERATE
BraTS20_Training_005	Pituitary Gland	22.0	5.4	0.617	MODERATE

BraTS20_Training_012	Pituitary Gland	32.0	3.1	0.736	MODERATE
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Rule Engine Thresholds

Parameter	HIGH Threshold	MODERATE Threshold
Distance to OAR	< 3.0 mm	-
AI Uncertainty	> 0.75	> 0.50
Tumor Volume	-	> 25.0 cc

This report is generated by a decision-support system and does not replace clinical judgment. Research prototype only. Not for clinical use. Data: BraTS 2020 (Kaggle). Guidelines: AAPM TG-132, ESTRO, QUANTEC, ASTRO, NRG Oncology.